Returns Wind Tunnel — Final Proposal

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Summary

Before a new shoe goes live, a team of agents tests it on virtual shoppers and on virtual AI shopping assistants, finds what would make it come back, and fixes the product page and product feed first. Fixes include fit notes, plain colour descriptions, a photo shot list that staff shoot (natural light, extra angles), and structured fit facts for AI buyers. People approve everything customers see, and real returns after launch recalibrate the tunnel. We estimate about 20K fewer returns a year, worth about $1.9M (base case), plus $0.5M–$2.0M of upside as AI assistants buy more of ALDO's online orders.

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1. Problem and thesis

Returns are written into the product page, and we learn too late. A shopper buys what the page tells her; when it says nothing about fit or the photo misleads on colour, the pair comes back, and only weeks of returns reveal why. ALDO e-commerce customers return about 327K pairs a year, worth $22.6M in refunds.

Return reason	Share of returned units
Change of mind*	38.2%
Too big	17.3%
Too small	16.9%
Logistics (lost in transit, return without item)	8.9%
Uncomfortable	6.4%
Not as expected / colour	4.6%
Quality / damage	3.9%
Wrong item	2.7%
Late delivery	0.8%

Size, colour, comfort and "not as expected" are about 46% of returns, and a better page can warn about all four. *"Change of mind" is overstated: it is 61% of store-counter returns, where staff pick

the code, but only about 10% when customers choose the reason online, where 58% of returns are size.

Why today's approach falls short:

- New styles launch with a generic size chart and studio photos.
- Fit and colour problems surface only after weeks of returns: a style needs a median of 37 days to collect 10 size returns.
- AI assistants (ChatGPT, Gemini) now buy from the product feed alone, which says nothing about fit.
- Nobody tests a page the way a shopper reads it before launch.

Thesis: if agents can shop a new style the way people and their AIs will, before launch, they can find and fix the returns hiding in the page: sales that stay sold from day one.

2. What ALDO's data tells us

1. **Size is the biggest fixable cause:** 35% of returns, $8.1M refunded per year.
2. **Fit errors point one way:** of 1,082 styles with 50+ size returns, 309 run clearly small and 356 clearly big; they hold 64% of size returns.
3. **A few styles drive most returns:** the top 10% of styles produce 73% of returns. Elliette returns 40% of 1,779 pairs; 70% of its size returns say "too small".
4. **"Change of mind" is often the wrong look:** when it ends in an exchange, 49% switch to another style.
5. **Customers show what they would have kept:** 4,615 style-to-style exchanges (Elliette → Braxtonn 10 times, Kamrynn 9 times).
6. **Colour and look surprises: about 15K returns a year** (2.9K name the colour, 12.1K "not what I expected"), across 1,374 styles; 91% come back by mail.

Sources: OMS returns (649,745 lines), OMS orders sample (1M lines), in-store exchanges of web orders (61,934), Feb 2024 – Jan 2026. Rates from the orders sample are estimates.

3. Our solution: a wind tunnel for every new shoe

Like a car tested in a wind tunnel before the road, each new style is shopped by virtual customers before it goes live.

1. **Test:** virtual shoppers and virtual AI assistants shop the new style.
2. **Diagnose:** the agents predict the return rate, with a range, and why.
3. **Fix:** description, fit note, colour text, photo shot list, AI-feed facts.
4. **Approve:** a merchandiser signs off; staff shoot the photos.

5. **Launch,** then compare with real returns.

The tunnel re-tests until the predicted return rate stops falling, and real returns recalibrate the virtual shoppers for the next launch. Existing high-return styles can go through the same tunnel with their current page.

4. Two groups in the tunnel: people and their AIs

More shoppers ask ChatGPT or Gemini to buy for them. Shopify Agentic Storefronts (January 2026, on by default for eligible stores) put products there, and ALDO is moving to Shopify. An AI buyer never sees the page.

	Group 1: virtual shoppers	Group 2: virtual AI assistants
Who	40 personas from ALDO's size mix, foot widths, brands worn and occasions	10 AI shoppers buying for a user profile ("wears Nike 8, wide feet")
What they see	The product page: photos, description, size chart, colour name	Only the product feed: title, attributes, size list, as Shopify sends to ChatGPT
How they go wrong	"Photo looks brighter than Rosewood." "Nothing says narrow toe, so I took my usual 8."	No fit data in the feed, so they order the user's usual size every time
What the agent fixes	Description and fit note; a photo shot list (natural light, side, on foot, colour close-up) for staff to shoot	Structured fit facts in the feed (runs_small: half, width: narrow, colour text) and a fit-question tool over MCP

The same shoe fails differently for a person and for an AI: a person is misled by a photo, an AI by missing data. One tunnel catches both, so ALDO is ready for AI buyers before they become a large share of orders.

5. Demo: the "Rosewood" boot

"Rosewood" is a fictional style; all numbers in this section are illustrative prototype output, not ALDO data.

Run 1, what the virtual customers said:

- Shopper, size 8, wide feet: "Took my usual 8. Nothing says the toe is narrow. Likely return: too small."
- Shopper, gift buyer: "I expected a rosy pink. The photo looks brighter than the name. Likely return: colour."
- Shopper, office wear: "Can't tell the heel height from the photos. Unsure about comfort."
- AI assistant, user wears Nike 8: "The feed lists sizes only. Ordering 8."

The agent's fixes:

- Fit note: "Narrow toe. If your feet are wide, size up half."
- Colour: "Rosewood: a deep red-brown with a slight purple cast. Darker than it looks on most screens."
- Shot list for staff: natural-light front and side, on foot, colour close-up with a white card, heel shown in profile.
- AI feed: runs_small: half, width: narrow, heel_cm: 6, colour text; MCP fit tool on.

Photos come back; the agent checks them with a vision model and re-runs the tunnel.

Group	Run 1	After fixes
Virtual shoppers	22% ± 6	14% ± 5
Virtual AI assistants	35% ± 9	12% ± 6

6. Why this is an agent, not code

A team of agents plans, imitates, diagnoses, fixes and checks.

Agent	What it does	Key tools
Director	Plans the test; decides how many runs, when to fix, when to stop, what to send to people	run_tunnel, request_approval
Persona builder	Creates shoppers and AI assistants from ALDO size, width and return-reason mixes	get_aldo_baselines
Virtual shoppers	Read the page and photos like a person, choose a size, say what they expect and what would make them return	vision, get_page
Virtual AI buyers	Buy from the product feed alone for a user profile	get_feed, MCP
Analyst	Groups the reasons, predicts the return rate with a range, compares with similar ALDO styles	cluster_reasons, similar_styles
Fixer	Drafts description, fit note, colour text, photo shot list and feed attributes	draft_edit, shot_list
Photo checker	Checks staff photos: right style and colour, natural light, white card	vision

What code alone can't do: read a photo and description the way a shopper would and predict the misunderstanding; pretend to be an AI assistant buying from a feed; write the fix in ALDO's voice and decide which fix matters most; decide when another run is worth it and what needs a person.

Each run is a loop: plan the test → act (simulate) → observe the reasons → reflect on the biggest cause → fix → test again. Different styles take different paths: a fit problem leads to a note, a colour problem to a photo task, a missing-data problem to feed attributes.

7. Human in the loop, uncertainty and learning

Human in the loop:

- The photo team shoots the agent's shot list. The agent never generates or edits product photos: shoppers only see real pairs.
- A merchandiser approves every page edit and colour description before launch.
- The e-commerce lead approves fit facts before they go into the feed for AI buyers.
- A person can override the tunnel's verdict at any time.

Uncertainty: every prediction comes with a range (22% ± 6%, not just 22%), based on how much the virtual customers disagree. A wide range makes the director run more customers before deciding; a fix that doesn't beat the range stops the editing; low confidence on fit makes the page ask shoppers their usual size instead of asserting one.

Learning:

- Backtest before trusting it: run the tunnel on old pages of styles with known returns (e.g. Elliette: 70% of size returns "too small"). Success means the tunnel names the same top reason. Backtest set: 662 ALDO styles with a clear size direction.
- After every launch, compare predicted with real returns at 30 and 60 days and re-weight the personas that were wrong. A style's first 5 size returns already show its true fit direction 91% of the time, so feedback is fast.

8. On AWS

- New-style pages, photos and feeds sit in S3; ALDO return data is queried with Athena.
- Bedrock AgentCore Runtime hosts the agents, built with Strands: Claude Sonnet for the director, persona builder, analyst and fixer; Claude Haiku for the 50 virtual customers, run in parallel; Claude vision for the photo checker.
- DynamoDB stores runs, verdicts and fixes; Bedrock Guardrails filter content.
- After approval, fit facts go into the Shopify product feed and an MCP fit tool on AgentCore Gateway, so AI buyers see the same truth as people.
- EventBridge brings real returns at 30 and 60 days back for calibration.
- CloudWatch + AgentCore Observability trace every run, persona answer, fix and approval. This reuses our team's Terraform stack in us-east-1.

9. Business impact

Base case: about 19.5K fewer returns a year (about 6% of the 327K), worth about $1.9M: $1.35M of refunds kept as sales and $0.6M of returns handling avoided.

What the tunnel fixes on the page	Base from ALDO data (per year)	Assumption	Returns avoided
Fit notes on styles that run small or big	72.2K size returns in skewed styles	20% prevented	14.4K
Colour: natural-light photos, colour text	15.0K colour + "not as expected" returns	20% prevented	3.0K
Comfort: toe shape and heel height made clear	20.9K "uncomfortable" returns	10% prevented	2.1K
Total			**19.5K**

Scenario	Returns avoided per year	Value per year
Conservative (half the rates)	10K	$1.0M
Base case	20K	$1.9M
Upside (×1.6)	31K	$3.1M

Value = refunds kept as sales ($69 average refund) + returns handling avoided (about $30 per footwear return, modelled). Rates are our assumptions, to be measured against styles launched without the tunnel. The $1.35M is revenue kept, not profit; its margin plus the avoided markdown on returned stock is the full gross-margin effect.

Growth upside, AI buyers (not in the base case): AI referrals to US retail sites grew about 7× in the 2025 holiday season and converted 31% better than other traffic (Adobe). If AI assistants place 5–20% of ALDO's online orders, accurate fit facts are worth an extra $0.5M–$2.0M a year: fewer AI returns, and AIs choosing ALDO because it can answer "which size?".

Sources: Adobe via Digital Transactions; McKinsey via Digital Commerce 360; Paz.ai, "UCP vs ACP".

10. Responsible AI

The agents may test and propose. People decide what customers see.

Area	Control
Honesty	Real photos only: the agent never generates or retouches product images. It writes shot lists; staff photograph real pairs. Descriptions state facts, never hype.

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Fairness and bias	Virtual customers differ by size, width, brands worn and use, drawn from ALDO aggregates, never by gender, ethnicity or age stereotypes.
Privacy	No real customer in the tunnel: personas are synthetic and built from anonymised totals. No individual's history is used or stored.
Explainability	Every fix shows its evidence ("18 of 40 shoppers took their usual size; 11 said nothing mentions a narrow toe"). Each run is logged with the persona answers behind it.
Authority	Nothing publishes itself: page edits, colour text and feed facts need human approval. The tunnel can't change prices, policy or the assortment, and can't cancel a launch.
Humility	Results are predictions, labelled as such: they carry ranges, are backtested on known styles and compared with real returns after launch. Personas that miss are re-weighted.